A Clinical Workflow Integration Layer for Tuberculosis Diagnosis in Chest X-ray Imaging

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Abstract—**Tuberculosis (TB) remains a leading infectious cause of death worldwide. Computer-aided detection (CAD) for TB on chest X-ray is now mature: the World Health Organization (WHO) recommended it for screening in 2021, and six commercial products met WHO performance standards in a 2025 policy update. Detection accuracy is strong, but the move from laboratory performance to reliable hospital deployment is incomplete. Workflow integration, input quality assurance, clinician-facing explanation of findings, and post-deployment monitoring remain underdeveloped in current systems. We present a prototype clinical workflow integration layer that wraps a mature TB CAD backbone. Its lead contribution is a clinically grounded multimodal explanation module that turns model findings into natural-language statements linking each suspected abnormality to its visual basis on the radiograph, a capability not clearly deployed in the commercial products we reviewed. Supporting modules add an input quality gate, a structured evidence interface, and a post-deployment monitoring dashboard. We frame the system as an academic prototype for hospital chest X-ray screening. Alongside the workflow layer, we develop and benchmark custom tuberculosis classification and localization models—including DraxNet and FlipR, a lightweight ResNet-18 variant with a bilateral-symmetry block—against standard ResNet and YOLO baselines on the TBX11K dataset.**

Keywords—Tuberculosis; Computer-Aided Detection; Chest X-Ray; Clinical Workflow; Medical Imaging; Computer Vision

I. Introduction

Tuberculosis (TB) is an airborne bacterial infection caused by *Mycobacterium tuberculosis* that primarily affects the lungs. It spreads when an infected person coughs, sneezes, or speaks, releasing bacteria that others may inhale. TB is both preventable and curable, yet it remains one of the leading infectious causes of death worldwide. The World Health Organization (WHO) estimated that 10.8 million people fell ill with TB in 2023, and approximately 1.25 million died from the disease [1]. Low- and middle-income countries bear a disproportionate share of this burden, often compounded by limited access to diagnostics in rural areas, delayed presentation, and overburdened public health infrastructure.

Chest X-ray (CXR) is the most widely used imaging modality for TB screening. A standard posteroanterior chest radiograph can reveal patterns associated with pulmonary TB, including upper-lobe infiltrates, cavitary lesions, pleural effusions, and lymphadenopathy. In screening settings, CXR serves as a triage tool: patients whose radiographs show findings suggestive of TB are referred for confirmatory bacteriological testing, typically sputum smear microscopy or molecular assays such as GeneXpert. This two-stage workflow, radiographic triage followed by laboratory confirmation, is central to national TB control programs in high-burden countries [2].

Interpreting chest radiographs for TB depends on trained readers. In many high-burden countries, radiologists are scarce, particularly in primary care facilities and rural health units where screening volumes are highest [2]. This bottleneck has driven the development of computer-aided detection (CAD) systems that use artificial intelligence (AI) to analyze CXR images and flag those suggestive of TB.

A. Maturity of TB CAD Systems

CAD for TB on chest X-ray is no longer an emerging technology. In 2021, WHO issued a policy recommendation supporting CAD as an alternative to human readers for TB screening and triage [3]. In June 2025, WHO reported that six commercial CAD products met WHO target performance standards for TB screening in individuals aged 15 years and older [4]. The WHO target product profile specifies minimum sensitivity of 90% and specificity of 70% for triage, benchmarks that multiple products now meet or exceed [5].

The broader evidence base reinforces this maturity. A systematic review of machine learning for chest X-ray interpretation found strong diagnostic performance across a range of thoracic conditions, though it also noted limited prospective validation and real-world deployment evidence [6]. A meta-analysis of CAD accuracy during active case finding for pulmonary TB in Africa reported pooled sensitivity of 0.87 and specificity of 0.74, which approach but do not uniformly meet the WHO screening target [35]. A 2025 meta-analysis of five commercial CAD products found individual sensitivities ranging from 86% to 91%, with some products meeting the WHO threshold and others falling slightly below depending on the operating point selected [36]. A 2024 reader study demonstrated that AI assistance raised physician diagnostic performance, measured by the area under the receiver operating characteristic curve (a summary statistic capturing accuracy across all decision thresholds), from 0.773 to 0.874 and helped non-radiologist physicians perform on par with radiologists [7]. In a real-world observational study, radiologists showed complete agreement with AI findings in 86.5% of cases, with only 0.5% of findings classified as missed by the AI [8].

These results indicate that the field has moved beyond asking whether AI can achieve useful diagnostic accuracy on chest X-rays. The question now is whether these systems can function reliably and safely when integrated into routine hospital workflows.

TABLE I. FEATURES PRESENT IN COMMERCIAL TB CAD PRODUCTS AND THE PROPOSED LAYER

Feature	**qXR** [18]	**CAD4TB+** [19]	**Lunit** [20]	**Annalise** [21]	**Siemens** [22]	**Proposed**
TB-specific support	✓	✓	✓			✓
Heatmap / overlay localization	✓	✓	✓	✓	✓	✓
Worklist / prioritization	✓		✓	✓		
Configurable threshold tuning		✓				✓
Monitoring / analytics dashboard	✓	✓	✓			✓
User-facing input quality gate					✓	✓
Natural-language finding explanation						✓

B. Related Work

Beyond deployed commercial systems, a substantial body of academic research has produced TB CXR analysis methods and prototypes. Early deep-learning work showed that convolutional neural networks could classify pulmonary TB on chest radiographs at accuracy comparable to radiologists [14], and subsequent work produced efficient architectures pairing fast screening with saliency-based visualization [15]. The TBX11K benchmark reframed TB CAD as a detection problem with region-level annotations and released reference detectors [44], [45], which the present work builds on while developing and benchmarking custom classification and localization models (DraxNet and FlipR) against standard ResNet and YOLO baselines. On the explanation side, research prototypes have begun coupling detectors or image encoders with language models: Danu et al. at Siemens Healthineers couple a thoracic abnormality detector with a domain-adapted large language model to generate chest-X-ray findings [12], while general medical vision–language models such as LLaVA-Med [13] and CXR-LLaVA [26] demonstrate image-conditioned radiology text generation, and recent surveys catalogue the rapid growth of vision–language models for medical report generation and visual question answering [27]. These remain research prototypes rather than deployed clinical features: trained weights are frequently unavailable, evaluation is typically retrospective, and image-conditioned generative models in particular are prone to hallucination [12], a risk that motivates the grounding-by-construction design adopted here.

Among deployed systems, we compared five commercial CAD products: qXR (Qure.ai), CAD4TB+ (Delft Imaging), Lunit INSIGHT CXR, Annalise Enterprise CXR, and Siemens AI-Rad Companion. As Table I summarizes, most provide TB detection, heatmap-based localization, and in some cases worklist prioritization and analytics. From official product documentation, two capabilities were not clearly present as deployed features:

- **Natural-language explanation of findings.** No reviewed product provides a deployed feature explaining diagnostic conclusions in clinician-facing causal language. As discussed above, the most directly relevant approach, Danu et al. at Siemens Healthineers [12], remains a research prototype not exposed as a documented feature in Siemens' commercial AI-Rad Companion product [22].
- **Integrated workflow.** No reviewed system combines input quality gating and natural-language explanation into a single layer.

The proposed system occupies this gap as a prototype demonstrating how such features could be integrated.

C. Gaps in Clinical Workflow Integration

Despite strong detection performance, reviews of CXR AI deployment consistently identify a gap between laboratory accuracy and clinical implementation. A review of practical AI in chest radiography found that only a small proportion of published studies are prospective or clinically embedded, and that workflow integration, local validation, and human factors remain under-studied [9]. The same body of literature argues that clinical expertise, usability, and implementation context remain essential even when model performance is strong [6], [9]. Real-world deployment studies reinforce this: an implementation of qXR for TB screening in India identified barriers including software installation, workflow integration, network connectivity, and limited human resources for interpreting findings [32]; a deployment study in Vietnam found only 47.3% concordance between physician and CAD interpretations at three months [33]; and early user interviews across multiple countries found that clinicians valued CAD most when it complemented rather than replaced clinical judgment [34]. Usability testing of a CXR AI tool in a simulated clinical workflow reported that users wanted clearer explanatory support and suggested the tool would be most useful in understaffed radiology departments, for trainees, and for non-radiologist physicians [10].

Several specific gaps emerge from this literature:

- **Input quality.** CAD systems are trained on curated datasets. In routine hospital use, images may be poorly positioned, clipped, rotated, underexposed, or photographed from analog film. Poor input quality can degrade performance, but most systems do not expose a user-facing quality assessment before inference [9], [11].
- **Explanation of findings.** Current commercial CAD products provide heatmap overlays showing image regions that influenced the prediction. A systematic review of explainability methods in medical imaging AI found that saliency-based approaches often show low overlap with expert-identified regions of interest and do not convey the contextual reasoning clinicians use during diagnosis [24]. Alternative explanation

approaches, including textual descriptions and case-based reasoning, have been proposed as more clinically meaningful [25]. Non-radiologists may benefit from natural-language explanations linking a conclusion to its visual evidence [12].

- **Post-deployment monitoring.** A deployed system's performance may change due to population shifts, equipment changes, or software updates. The U.S. Food and Drug Administration (FDA) has identified data drift, concept drift, and model drift as risks to AI-enabled medical devices and has initiated research programs on proactive post-market monitoring [28]. Reviews note the need for ongoing monitoring, but this remains an operational gap in practice [9].

D. Scope and Contribution

This paper addresses the workflow integration gap by presenting a prototype **clinical workflow integration layer** for TB CAD in hospital CXR screening. The system wraps clinician-facing modules around a classification backbone:

1) A **multimodal explanation module** that generates natural-language descriptions of AI-detected findings, linking suspected abnormalities to localized visual evidence. This is the lead contribution.
2) An **input quality gate** that screens radiographs for acquisition defects before inference.
3) A **structured evidence interface** presenting the TB score, risk classification, quality status, and localized evidence in a compact clinical panel.
4) A **post-deployment monitoring dashboard** tracking system behavior over time.

The primary claim is clinical utility and feasibility: that this combination addresses documented needs in hospital TB screening. As a secondary contribution, we develop and benchmark two custom tuberculosis models—DraxNet and FlipR—for image-level classification and bounding-box localization, evaluated against standard ResNet and YOLO baselines.

The system addresses four design goals, each corresponding to a documented failure mode or usability gap:

1) **Reject or flag unsafe inputs before inference.** Prevent the classifier from operating on images likely to produce unreliable predictions.
2) **Provide clinically grounded explanations.** Translate model evidence into natural-language statements describing what was found, where, and why it matters clinically.
3) **Present evidence in a workflow-friendly format.** Consolidate risk score, quality status, heatmap, and explanation into a single panel for rapid triage review.
4) **Monitor deployment quality.** Track operational metrics to detect degradation or drift after hospital deployment.

II. SYSTEM ARCHITECTURE

The system processes a chest X-ray through a sequential pipeline:

1) **Image ingestion.** A digital chest radiograph (DICOM or standard image format) is received from the hospital imaging system or uploaded directly.
2) **Input quality assessment.** The image is evaluated for acquisition quality before diagnostic inference.
3) **TB classification.** If the image passes quality screening, it is processed by the TB scoring module, producing a continuous probability score and spatial activation maps.
4) **Multimodal explanation generation.** For classified cases, the system generates a natural-language explanation linking the predicted finding to localized visual evidence.
5) **Clinical interface rendering.** All outputs are presented in a structured evidence panel.
6) **Monitoring and logging.** Each case and its metadata are logged for the monitoring dashboard.

The remaining sections describe each component in turn: input quality assessment (Section III), tuberculosis classification (Section IV), tuberculosis localization (Section V), and the multimodal explanation module (Section VI), followed by two non-evaluated components, the structured evidence interface (Section VII) and the post-deployment monitoring dashboard (Section VIII). Each evaluated module is presented as a self-contained unit with its own dataset, preprocessing, architecture, training, and results.

III. INPUT QUALITY ASSESSMENT

The input quality gate evaluates each radiograph before it reaches the classifier. It assigns one of four status labels:

- **Acceptable:** meets acquisition criteria; proceeds to classification.
- **Borderline:** minor quality issues (slight rotation, suboptimal exposure) that may reduce confidence. Classification proceeds with a quality warning attached.
- **Repeat recommended:** significant defects (clipped lung fields, severe underexposure) likely to compromise reliability. The system recommends re-acquisition.
- **Manual review required:** non-standard input (photographed analog film, lateral view, non-chest image). The case is routed to human review.

The gate is a hybrid pipeline combining DICOM header checks, a learned view classifier, and an anatomical-segmentation-based sanity check. Concretely: (i) DICOM tags (`ViewPosition`, `BodyPartExamined`, `PhotometricInterpretation`) are read first; any study with a non-frontal `ViewPosition` is routed to manual review, and inconsistent `BodyPartExamined` triggers a rejection. (ii) For cases lacking reliable DICOM headers, a small view classifier trained on frontal / lateral labels from PadChest [48] serves as a fallback. (iii) The image is passed through the pretrained anatomical segmentation network from the `torchxrayvision` library [47], which produces per-pixel masks for left lung, right lung, heart, and other thoracic structures. The gate computes three summary statistics from these masks: the lung-area-to-image-area ratio (a field-of-view score, flagging clipped fields), the lateral offset of the lung centroid relative to image center (a rotation proxy), and the presence of both lung masks above a minimum area (a non-chest-image check). (iv) Classical image statistics (Laplacian variance for blur; 1st/99th percentile intensity gaps for under- or over-exposure) are thresholded on development data and added to the rule set. The combined rules map each image to one of the four status labels above.

This design is motivated by evidence that CXR AI performance degrades when applied to images differing from the training distribution. A systematic review of cross-population domain shift found that 81% of deep learning algorithms in radiology exhibited decreased accuracy on external datasets [37], and shortcut learning, where models rely on acquisition artifacts rather than pathological features, has been documented as a failure mode in CXR AI [11], [9]. The gate is engineered rather than novel; it is cited as prior art that a clinically adequate CXR screening system has a well-defined acquisition-quality model upstream of inference.

Local threshold calibration matters for adapting CAD to site-specific prevalence and clinical context. WHO has published calibration guidance [16], and CAD4TB+ offers dynamic threshold adjustment [19]. A 2025 triage study demonstrated the sensitivity–specificity tradeoff from threshold selection [17]. The classifier operating point is therefore retained as a configurable parameter, but it is not a headline contribution; it addresses a partially solved problem for which guidance already exists.

A. Results

IV. Tuberculosis Classification

The tuberculosis classification module is the diagnostic backbone: it assigns an image-level label to each chest radiograph. We treat it as a three-class problem—*healthy*, *sick (non-TB)*, and *TB*—following the label structure of TBX11K [44]. We study two standard convolutional baselines (ResNet-18 and ResNet-50 [53]) and two custom architectures (DraxNet and FlipR), and evaluate them against the WHO finding that multiple TB CAD products already meet screening performance standards [4].

A. Dataset

All models are trained on **TBX11K**, a public dataset of 11,200 chest radiographs with image-level class labels (*healthy*, *sick non-TB*, *TB*) and expert-annotated bounding boxes for active-TB and latent-TB regions [44]. We use the simplified, classification-oriented redistribution of TBX11K, the `tbx11k-simplified` release.[1] TBX11K was selected because, as of writing, it is one of the few publicly available TB CXR datasets that simultaneously satisfies three requirements for this work. First, it provides a large TB-specific sample, enough to train deep networks from scratch, whereas most public TB datasets contain only tens to hundreds of images. Second, it supplies expert bounding-box annotations for TB regions, which enable supervised object detection rather than post-hoc saliency maps; radiologist user studies report that heatmap-style saliency overlays are less intuitive and harder to read than localized, discrete findings [38], so region-level supervision is preferable for a clinician-facing system. Third, its demographic composition and its healthy, sick non-TB, and TB proportions (including the active-to-latent TB ratio) are constructed to approximate the class imbalance of real-world clinical screening rather than an artificially balanced split [45], so measured performance better reflects expected deployment behavior. The explicit *sick non-TB* class is important to this realism: in deployment a radiologist receives scans from the full patient stream, many of whom are sick with conditions other than TB, so a clinically useful model must separate TB from other disease rather than only from healthy lungs. A dataset of only healthy and TB images would overstate performance by omitting the non-TB pathology that drives most false positives in practice. It also supports a reproducible public train/validation split. The official TBX11K test labels are withheld for an ongoing online challenge [45], so we use the publicly available TBX11K validation set as our held-out test set for all experiments. Our results should therefore not be compared directly to those reported for SymFormer or other entries on the official TBX11K benchmark, which are measured on the confidential test set. The image-level labels are used for the classification experiments in this section; the same dataset's bounding-box annotations are used for the localization experiments of Section V.

B. Data Preprocessing

Each radiograph is resized so that its longest side is at most 512 pixels and then zero-padded to a fixed 512×512 input, preserving aspect ratio, and normalized with the ImageNet channel mean and standard deviation.

The ResNet-18, ResNet-50, and DraxNet models are trained without data augmentation. This choice is deliberate: unlike natural-image object detection, screening chest radiographs are acquired under a standardized posteroanterior protocol and are largely uniform in pose and framing, so aggressive augmentation risks introducing distribution shift rather than useful invariance. FlipR is the exception; it is trained with a deliberately low-intensity augmentation pipeline (Albumentations) applied to the training split only, while the validation split uses resize, pad, and normalization alone. The FlipR training augmentations are: horizontal flip ($p = 0.5$); a small affine transform (translation $\pm 5\%$, scale 0.9–1.1, rotation $\pm 10°$, $p = 0.5$); mild random brightness/contrast jitter (± 0.15, $p = 0.5$); and Gaussian noise ($p = 0.2$), followed by ImageNet normalization. The low transformation strengths reflect the expectation of near-uniform CXR input rather than the wide pose variation typical of general object-detection datasets.

C. Model Architecture

We evaluate four image-level classifiers. Two are standard convolutional baselines, ResNet-18 and ResNet-50 [53], which provide well-understood reference points at 11.7M and 25.6M parameters respectively. The remaining two are custom architectures.

DraxNet is a custom convolutional classifier (17.0M parameters).

FlipR is a lightweight custom model (2.2M parameters) built on a ResNet-18 backbone. Its principal modification is

[1] https://www.kaggle.com/datasets/vbookshelf/tbx11k-simplified

a new block, also named *FlipR*, inserted between backbone stages. The block is partially inspired by the SymAttention mechanism of the "Revisiting Computer-Aided Tuberculosis Diagnosis" work [45], which observes that *bilateral asymmetry* in the chest radiograph is a key radiographic indicator of TB. The FlipR block computes the difference between each feature map and its horizontally flipped counterpart—the flipped version first smoothed with average pooling to tolerate imperfect anatomical symmetry—and from this difference derives a gated mask, in the manner of a gated linear unit (GLU), that is broadcast across all channels at once. The design goal is to test whether a simple, parameter-efficient symmetry-aware gate can capture the same bilateral-asymmetry cue as heavier spatial-transformer approaches; FlipR has roughly one-fifth the parameters of ResNet-18.

D. Model Training

All four classifiers are trained from scratch, with no ImageNet pretraining, so that the comparison reflects each architecture's inductive bias rather than transferred features. All models are implemented and trained in PyTorch, with a fixed random seed of 42 throughout for reproducibility. Training uses a batch size of 16, a fixed 512×512 input resolution, and 100 epochs. Data augmentation is applied only to FlipR, as described above; the ResNet baselines and DraxNet are trained on unaugmented inputs.

E. Results

Table II reports image-level classification performance on the held-out TBX11K validation set. All models were trained from scratch without pretraining. Among the models with completed runs, DraxNet achieves the strongest aggregate results, with the highest TB-class AUC (0.9990), accuracy (98.97%), and macro F1 (97.98%). This aggregate advantage is driven by near-perfect specificity rather than by improved TB detection: DraxNet reaches a TB-class specificity of 99.91% but a TB-class sensitivity of 91.67%, slightly below the 92.50% of ResNet-18. All three models share the same error profile, pairing TB specificity above 99% with TB sensitivity between 87.5% and 92.5%. The models are therefore conservative, rarely raising false alarms but missing 7.5 to 12.5% of TB cases. For a screening application the missed-case direction is the costly one, so the aggregate metrics overstate the operational readiness of all three models. Read against the WHO target product profile for TB triage, which specifies a minimum sensitivity of 90% and specificity of 70% [5], the clinical picture is asymmetric: all three models clear the specificity floor by roughly 30 percentage points, but their TB sensitivity (87.5 to 92.5%) only borders the 90% sensitivity floor and falls below it for ResNet-50. This high-specificity, lower-sensitivity regime is not an artifact of our architectures. The TBX11K reference method SymFormer likewise operates at a conservative point (a specificity of 97.3% for its Deformable DETR variant), and the original benchmark explicitly frames high specificity as the desirable behavior, identifying the principal failure mode as baseline detectors that over-predict TB and collapse to near-zero specificity [45]. The asymmetry therefore reflects the TBX11K task itself rather than a defect of our models, although the absolute values remain not directly comparable given the different split and label set. Clinically, this matters because the two error types are not equivalent. A missed TB case (false negative) delays treatment and allows continued community transmission, the outcome a screening program most needs to avoid, whereas the very low false-positive rate limits unnecessary confirmatory sputum or GeneXpert testing and reduces patient burden in resource-limited settings. The conservative operating point is therefore well suited to a confirmatory, radiologist-in-the-loop role that prioritizes specificity, consistent with the positioning of the explanation module in Section VI, but it is not yet safe as an autonomous rule-out screener; closing the residual sensitivity gap, for example by threshold recalibration toward higher sensitivity, would be required before frontline screening use. ResNet-18 slightly outperforms the larger ResNet-50, and the gap is widest on TB sensitivity (92.50% versus 87.50%). We hypothesize that this inversion of the usual expectation that more parameters yield better performance arises from three compounding conditions specific to this setting: the models are trained from scratch without ImageNet initialization, the dataset is modest in scale, and the TB class is the minority. Under these conditions the additional capacity of ResNet-50 is more readily spent overfitting the majority healthy and sick non-TB classes than learning discriminative TB features, so the degradation concentrates on minority-class recall (sensitivity) rather than on specificity or overall accuracy. The lower-capacity ResNet-18, together with the compact custom models, appears to act as an implicit regularizer at this data scale. This favors compact architectures for the intended deployment. TB screening is concentrated in high-burden, resource-limited settings where inference frequently runs on modest CPU-only or edge hardware without a dedicated GPU, so a model such as DraxNet (17.0M) or FlipR (2.2M) that matches or exceeds ResNet-50 (25.6M) at a fraction of the parameter count and compute is preferable on both accuracy and operational grounds. Accuracy for every model exceeds the 84.8% reported for radiologists on TBX11K [45], though, as noted in Section IV, the validation-set protocol and the merged three-class label set make these figures not directly comparable to the official benchmark. FlipR, the lightweight symmetry-aware model, has only 2.2M parameters, roughly one-fifth of ResNet-18.

V. Tuberculosis Localization (Object Detection)

Image-level classification answers *whether* a radiograph is suggestive of TB but not *where*. The localization module complements it by predicting bounding boxes around TB-related regions, which also supplies the region-level evidence consumed downstream by the multimodal explanation module (Section VI).

A. Dataset

Localization uses the same TBX11K data [44], here with the region-level bounding-box annotations rather than the image-level labels. As in Section IV, the official TBX11K test labels are withheld for an ongoing online challenge [45], so detection is evaluated on the publicly available TBX11K validation set used as the held-out test set, and the reported numbers are not directly comparable to SymFormer or other entries on the official TBX11K detection benchmark. Two foreground classes are predicted: active TB (ATB) and latent TB

TABLE II. TUBERCULOSIS CLASSIFICATION RESULTS

Model	AUC_{TB}	Acc.	Sens.	Spec.	Avg. Prec.	Avg. Rec.	Avg. F1	Params
ResNet-18	0.9973	98.41	**92.50**	99.39	97.27	96.86	97.06	11.7M
ResNet-50	0.9965	97.86	87.50	99.47	96.99	95.13	96.01	25.6M
DraxNet	**0.9990**	**98.97**	91.67	**99.91**	**99.01**	**97.05**	**97.98**	17.0M
FlipR	—	—	—	—	—	—	—	**2.2M**

AUC_{TB} is the area under the ROC curve for the TB class. Accuracy, sensitivity, specificity, and the averaged precision, recall, and F1 are reported as percentages. Sensitivity is recall for the TB class; specificity is recall for the non-TB class (healthy and sick non-TB combined). Avg. Prec., Avg. Rec., and Avg. F1 are precision, recall, and F1 macro-averaged across the three dataset classes.

(LTB), referred to in TBX11K as *ActiveTuberculosis* and *ObsoletePulmonaryTuberculosis*, respectively. Obsolete pulmonary (latent) TB denotes the radiographic residue of prior, healed or inactive infection, such as fibrotic bands, calcified granulomas, and apical scarring, that is not contagious and does not require immediate treatment. Active TB, by contrast, denotes ongoing infection that is transmissible and clinically urgent. The two classes therefore differ not only in radiographic appearance but in clinical priority, which motivates reporting detection performance per class rather than only in aggregate. The active and latent designations are assigned at the bounding-box level rather than the image level: a CXR carrying an image-level TB label may contain only active-TB lesions, only latent-TB lesions, or lesions of both types on the same radiograph. In TBX11K this corresponds to 924 images with active TB only, 212 with latent TB only, and 54 with both [45]. This box-level, mixed-type annotation is consistent with real-world conditions, where a single patient's radiograph can simultaneously show active disease and healed or inactive scarring. The detected region centroids are intersected with a pretrained anatomical lung-zone segmentation [47] to obtain an anatomical descriptor (lung side and zone) that the explanation module of Section VI consumes.

B. Model Architecture

We evaluate the YOLO26 single-stage object detector and a variant, YOLO26 with a DraxNet backbone, in which the detector's default feature extractor is replaced by the DraxNet backbone used in the classification experiments. This isolates the effect of the backbone on TB localization while holding the detection head and training protocol fixed.

C. Model Training

Both detectors are implemented and trained in PyTorch, with the same fixed random seed of 42, a batch size of 16, a 512×512 input resolution, and 100 epochs, matching the classification protocol for comparability.

D. Evaluation Metric

Detection performance is reported using mean average precision (mAP) over the bounding-box predictions, following the COCO benchmark [54] and the TBX11K benchmark [45]. We report two operating points: mAP_{50}, the mAP at a single intersection-over-union (IoU) threshold of 0.50; and $\mathrm{mAP}_{50\text{-}95}$, the mAP averaged over IoU thresholds from 0.50 to 0.95 in steps of 0.05.

E. Results

Table III reports per-class bounding-box localization performance. These figures are computed over the whole evaluation set for a self-contained assessment of the detector, but in the deployed system the localization model is run only on radiographs that the classification module of Section IV labels as TB at the image level; detections on images classified as non-TB are not produced. This cascaded use follows the TBX11K design, in which the image-classification result filters the detector and detected TB areas are discarded when a CXR is classified as non-TB [45], on the principle that lesion-level localization is meaningful only once an image is judged TB-positive. Replacing the YOLO26 backbone with the DraxNet backbone improves ATB localization substantially, raising $\mathrm{mAP}_{50\text{-}95}$ from 0.2110 to 0.2748 ($+0.0638$), but slightly reduces LTB from 0.0410 to 0.0310. The same backbone substitution raises overall detection $\mathrm{mAP}_{50\text{-}95}$ from 0.119 to 0.152 (a 28% relative gain) and mAP_{50} from 0.274 to 0.343. This gain is driven almost entirely by precision, which rises from 0.74 to 0.82, while recall is unchanged at approximately 0.26. The strong classification feature extractor therefore yields cleaner box proposals for the salient active-disease class without recovering more regions overall. Recall, not precision, is the bottleneck: both detectors localize confidently but retrieve only about a quarter of the annotated TB regions. Localization is consequently most reliable as supporting region-level evidence for the high-sensitivity classifier and the downstream explanation module of Section VI, rather than as a standalone detector. The very low absolute $\mathrm{mAP}_{50\text{-}95}$ for LTB, which the DraxNet backbone slightly worsens, is consistent with the severe region-level class imbalance in TBX11K (active-TB regions outnumber latent-TB regions by roughly four to one) and with the subtle, diffuse appearance of latent-TB lesions; the discriminative backbone appears to specialize toward the dominant active-TB class. Clinically, the high-precision, low-recall profile defines how localization should be used at the point of care: a flagged region is trustworthy and warrants focused radiologist review (a rule-in cue), but the detector must not be used to exclude TB, since most annotated lesions are not retrieved. The near-blindness to latent (obsolete) TB is less clinically severe than the aggregate mAP suggests. Obsolete pulmonary TB denotes healed, inactive disease that is neither transmissible nor immediately treatment-urgent, whereas active TB is the contagious, time-critical class that drives isolation and treatment decisions. Concentrating the model's localization capacity on active TB, even at the cost of latent-TB mAP and of lower aggregate scores, therefore aligns with clinical triage priorities rather than working against them. Used this way, the localization module functions as an attention-directing aid that grounds the natural-language output of the explanation module (Section VI) in a specific anatomical region, not as an independent diagnostic detector.

TABLE III. TUBERCULOSIS LOCALIZATION RESULTS

Model	Class	mAP$_{50}$	mAP$_{50\text{-}95}$
YOLO26	ATB	—	0.2110
YOLO26	LTB	—	**0.0410**
YOLO26 + DraxNet	ATB	—	**0.2748**
YOLO26 + DraxNet	LTB	—	0.0310

ATB = active TB; LTB = latent TB.

VI. Multimodal Explanation

The multimodal explanation module is the lead contribution. It translates the classifier's evidence into a natural-language statement that a clinician can read, understand, and act on without interpreting heatmaps or understanding the model's internals. Evidence for the value of natural-language explanation in CXR AI comes from user studies and clinician-interview research, though the demand picture is mixed. In a user-centered evaluation of explainability techniques on chest radiographs with 26 clinicians at a National Health Service trust in the United Kingdom, a majority stated that descriptive sentences should be added alongside visual explanations, and roughly half reported that colored heatmap overlays hindered readability [38]. A multisite prospective study of 220 physicians reading chest radiographs found that the form of AI explanation materially affected diagnostic performance and physician trust, indicating that explanation modality, not only the presence of an explanation, shapes clinical use [39]. Broader interview-based studies of AI decision support in medicine have surfaced consistent requests for contextual reasoning and explanatory information rather than predictions alone [40], [41]. In the TB CAD setting specifically, a qualitative study of radiologists using a commercial TB CXR product found opinion divided: one participant reported struggling to understand why the system reached a given conclusion, while another characterized the tool as an engine whose internal workings were not clinically relevant to them [42]. The module is therefore positioned as most useful for a radiologist-in-the-loop or confirmatory reader reviewing flagged cases, rather than as a universally demanded feature for frontline screeners who primarily act on the abnormality score.

Given a radiograph and the classifier's output, the module generates a statement identifying the suspected finding (for example, a cavitary lesion, upper-lobe infiltrate, or pleural effusion), its anatomical location (for example, the right upper lobe or left costophrenic angle), and a clinical reasoning link connecting the finding to the TB classification. An example output: *"TB suspected. The model identified an area of increased opacity in the right upper lobe consistent with an infiltrate. Upper-lobe infiltrates are among the most common radiographic findings in active pulmonary tuberculosis."* This differs from heatmap-only explanations in two ways. First, it names the finding in clinical vocabulary rather than highlighting a region. Second, it provides a reason, grounded in radiographic knowledge, for why that finding supports a TB classification.

VII. Structured Evidence Interface

All outputs are consolidated into a single panel for rapid clinical review: the TB probability score as a numeric value and color-coded risk band (low, moderate, high); the quality status with warnings for detected issues; a heatmap overlay showing regions influencing the prediction; and the natural-language explanation from the multimodal module. The panel uses a traffic-light color scheme for risk and quality status and is designed to be reviewable in under 30 seconds during routine triage.

VIII. Post-Deployment Monitoring

A monitoring dashboard tracks operational metrics to detect behavioral changes after deployment, drawing on the literature on post-deployment medical-AI monitoring [49], [50], [51], [29] and metrics published by existing TB CAD products [18], [19], [20]. Metrics are organized into five groups: input-side (case volume, quality-gate rejection rate, scanner/site/demographic mix relative to training); prediction-side (TB-score distribution, flag rate, per-finding trajectory, calibration metrics); outcome-side (radiologist override rate, CAD–human discordance, and bacteriologically confirmed TB yield among CAD-positive cases, explicitly called for in WHO guidance [2]); drift (Population Stability Index and Kolmogorov–Smirnov statistics on score distributions, with PSI < 0.1 stable, 0.1–0.25 flagged, ≥ 0.25 a major shift [50], plus higher-dimensional embedding drift detection [51], [52]); and fairness (sensitivity, specificity, and positive predictive value stratified by clinically relevant subgroups with per-subgroup confidence intervals, as required by regulatory guidance [28]). The dashboard does not autonomously adjust system behavior; it enables human intervention when metrics indicate degradation or drift [9], [29].

IX. Discussion

The modules form a single screening workflow rather than a set of independent models. The classification and localization experiments suggest that compact custom backbones can match or exceed standard ResNet baselines on TBX11K, while the explanation, interface, and monitoring components target the laboratory-to-deployment gaps identified in the Introduction. The intended contribution is integrative: each module is individually modest, but their composition addresses needs that detection accuracy alone does not. These claims are preliminary. The evaluation reported here is partial, and the end-to-end workflow argument is made rather than yet demonstrated in a clinical setting.

A. Limitations

This work has several limitations. First, the system has not been evaluated in a live clinical environment; evaluation relies on retrospective data and simulated workflows. Second, the explanation module's outputs have not been validated against expert radiologist interpretations at scale; incorrect explanations could reduce rather than build trust. Third, quality gate thresholds were set on development data and may require recalibration for different imaging environments. Fourth, the TB detection module processes only a single frontal (posteroanterior or anteroposterior) chest radiograph per case, a constraint shared by all WHO-approved TB CAD products, which are trained exclusively on frontal projections. Unlike a clinical PACS hanging protocol that presents current and prior images side by side, the system cannot compare serial radiographs. This restricts its use to screening of new patients at a single time point; it cannot track radiographic change over the course of TB treatment or detect progression and regression across follow-up examinations. Finally, the multimodal explanation module is a prototype, not a clinically validated explanation system: the grounding-by-construction approach reduces but does not eliminate error, since templated sentences can still propagate incorrect detector outputs and the LLM smoothing step could paraphrase findings in ways that shift clinical meaning, which is why the post-hoc verification step is included.

B. Future Work

- **Prospective clinical evaluation.** Hospital deployment with measurement of diagnostic outcomes, clinician behavior, and patient impact.
- **Cost-effectiveness analysis.** Assessment of whether the clinical workflow integration layer reduces unnecessary testing, missed cases, or clinical time. Cost data for AI-CAD workflows in high-burden settings are fragmented; a local pilot study would be needed to produce credible estimates.
- **Photographed-film workflows.** Extension of the quality gate to handle smartphone-captured analog radiographs, common in rural health facilities in high-burden countries, with dedicated evaluation of domain shift [23].
- **Formal explanation validation.** Systematic comparison of generated explanations against radiologist reports.
- **Trainee-oriented evaluation.** Primary user research with residents and non-radiologist physicians to determine whether a dedicated trainee mode is actually demanded, what form it should take, and whether it produces durable skill gains rather than in-session dependence.

X. Conclusion

TB CAD classifiers are now accurate enough for screening; the open problem is making their outputs safe, interpretable, and usable in hospital workflows. This paper presented a prototype clinical workflow integration layer that wraps a mature TB CAD backbone with a clinically grounded multimodal explanation module, an input quality gate, a structured evidence interface, and a post-deployment monitoring dashboard. A full evaluation of feasibility and clinical utility is left to future work.

Declaration on Generative AI

Generative AI tools were used to assist in drafting and refining language during manuscript preparation. The conception of the research, development of methodology, and all conclusions are the work of the authors.

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